



BITS Pilani

Cloud Computing

SEWP ZG527

AGENDA



“

Build scalable, secure and reliable solutions in the cloud



01 RECAP

- What is IaaS
- Introduce AWS
- AWS Reference Model



02 INFRASTRUCTURE AS A SERVICE



AWS Compute Services

- Instances, Clusters, VPC & Security Groups



AWS Storage

- File, Block & Object Storage



Cloud Data Services

- RDS, NoSQL, Data Storage & Analytics



Big Data Services

- HDFS, EMR & Data warehousing overview



SECURE

Enterprise-grade



SCALABLE

Scale up or down



RELIABLE

Highly available



COST EFFECTIVE

Pay only for what



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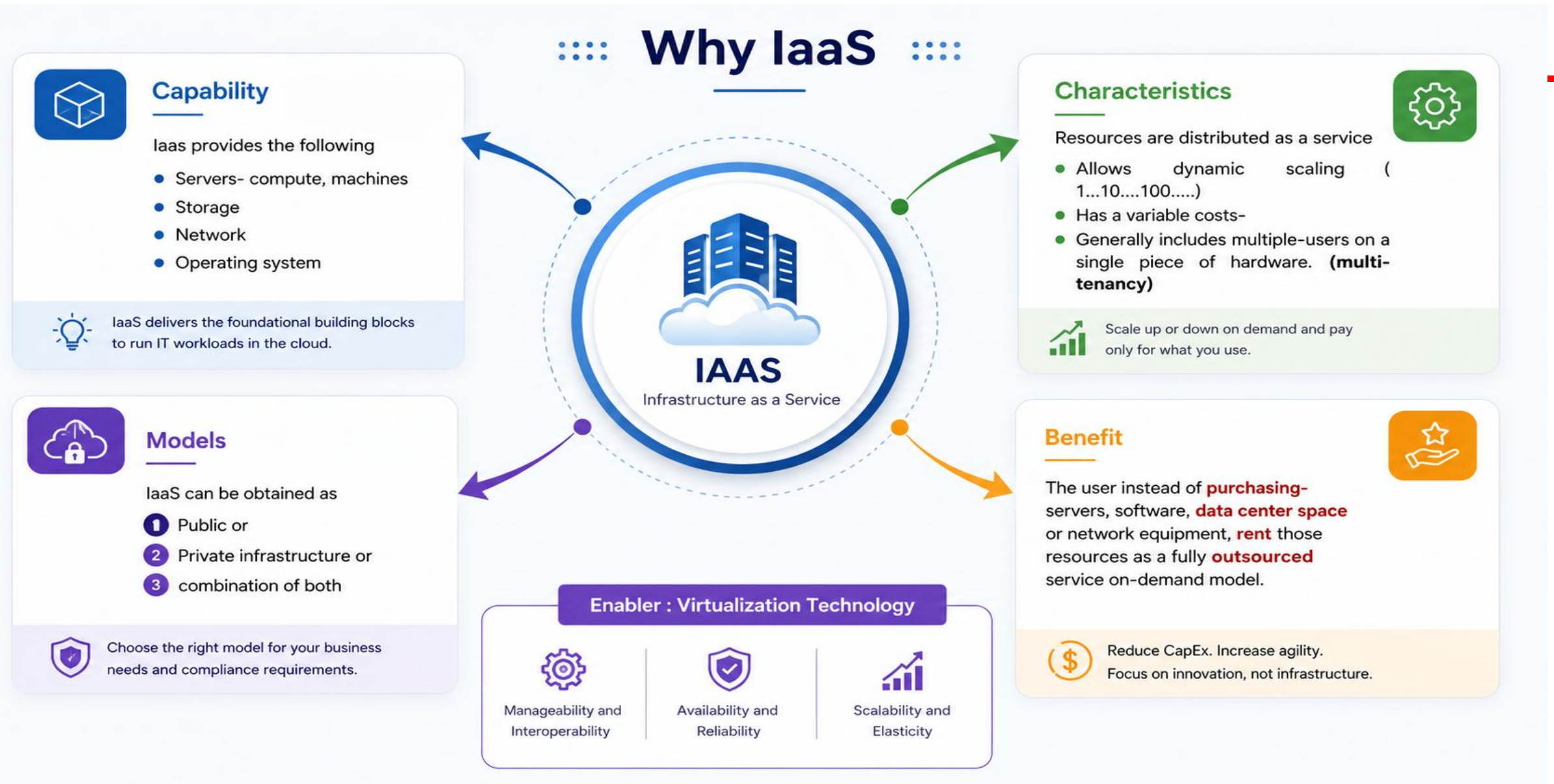
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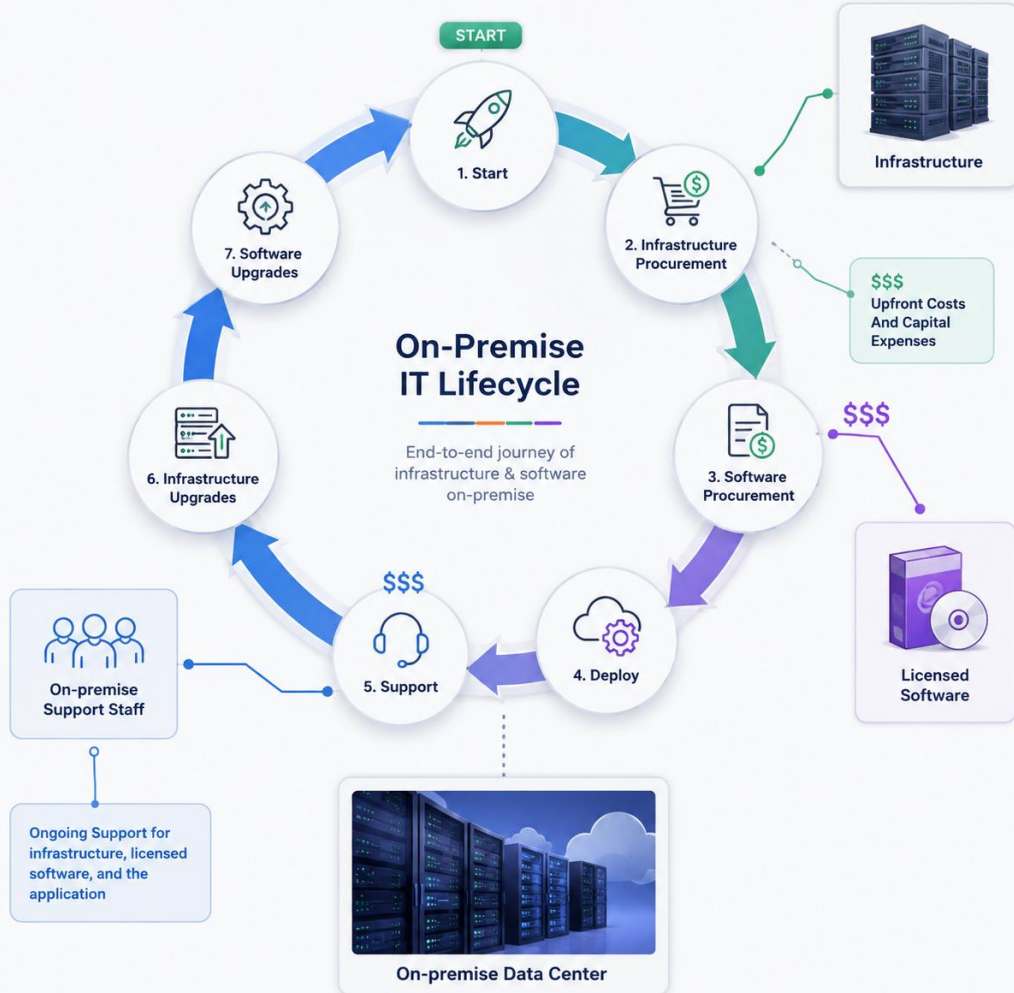
Recap



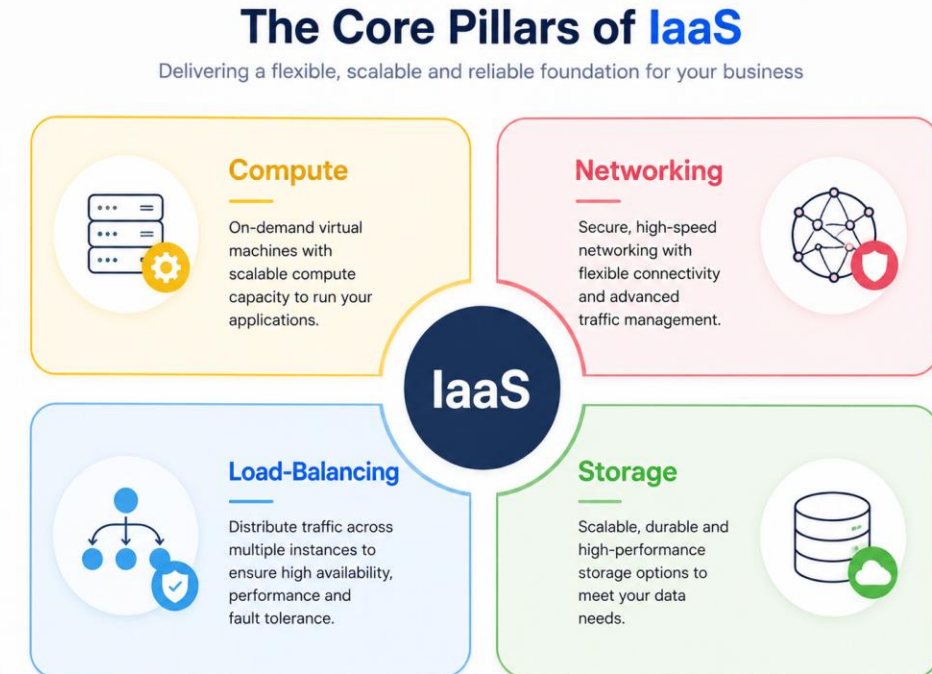
Infrastructure as a Service - Recap



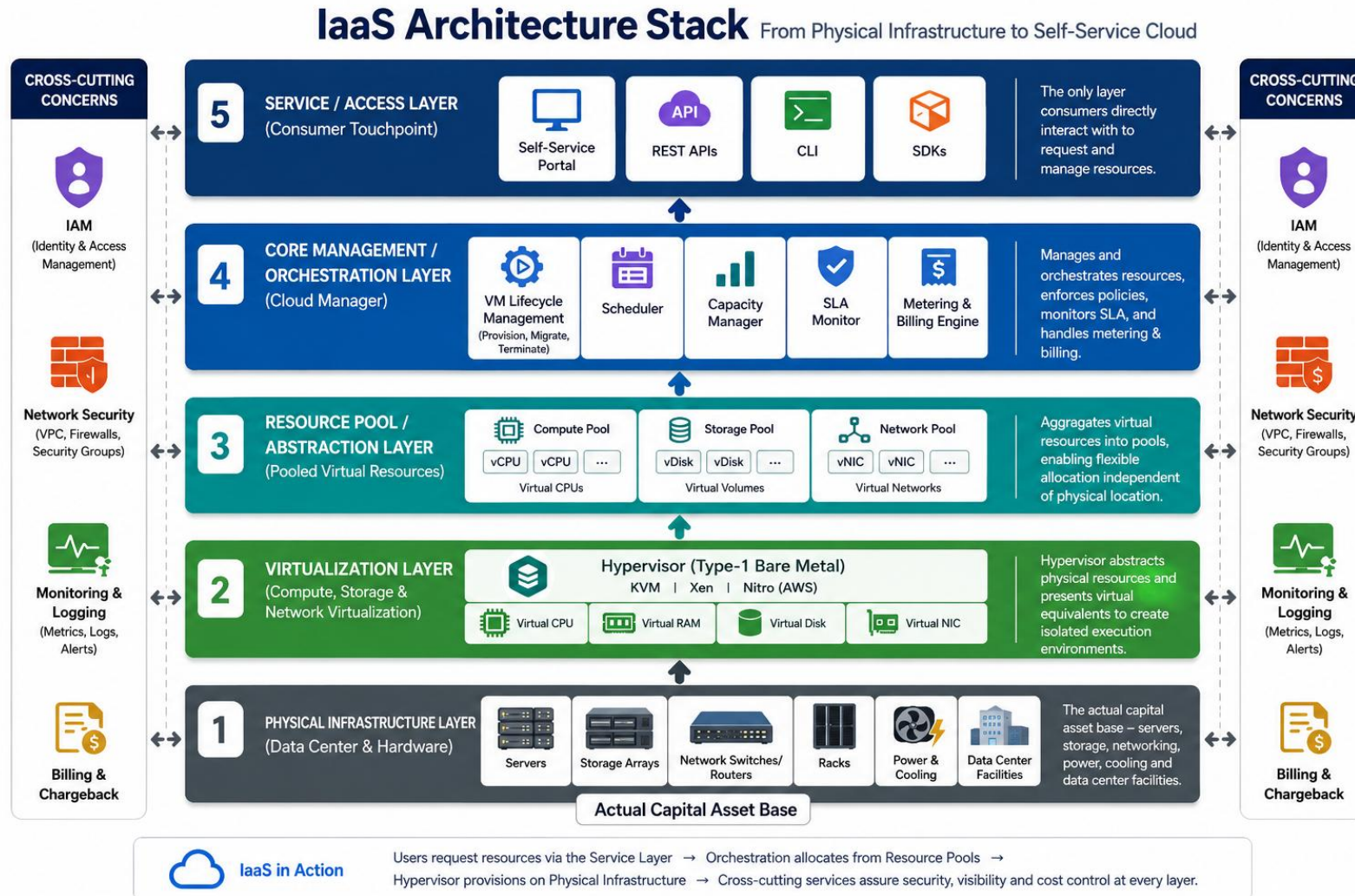
IaaS - Motivations



- No upfront cost
- Multi-tier auto provisioning within hours
- Scalability, services/resources are provided on demand
- Flexible/Secure direct connectivity
- SLA with 99.9% uptime



IAAS Architecture Model



This figure shows how the various technologies we learnt come together.

The final outcome of the interactions is what we colloquially refer to IaaS



innovate



achieve



lead

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IAAS – COMPUTE SERVICES



Instances



Clusters



VPC



Security Groups

EC2 Introduction

- Amazon EC2 is **AWS primary web service** that provides **resizable compute capacity** in the **cloud**.
- **Compute** refers to the amount of **computational power required to fulfill your workload**.
- Amazon EC2 allows you to acquire compute through the **launching of virtual servers** called **instances**.
- When you launch an **instance**, you can make use of the compute as you wish, just as you would with an on-premises server.
- **Users pay** for the **computing power** of the instance. Charged per hour while the instance is running. When you **stop the instance**, you are **no longer charged for EC2**, however **other services are still chargeable**.

NETFLIX



Amazon EC2



- Virtual Compute Cloud
- Root-level System Access
- Elastic Capacity
- Management API
- Scale in Minutes
- Multiple Instance Sizes
- Network Security Model

EC2 - Instance Type & AMI

- There are two concepts that are key to launching instances on AWS:
- (1) **Instance Type**: The amount of virtual hardware dedicated to the instance and
- (2) **AMI**: The software loaded on the instance.
 - AMI → Amazon Machine Image
- Note
 - Instance Type is similar to the processor*
 - AMI is similar to the OS*

AWS EC2 Instance Families at a Glance

Type	Family	Description	Mnemonic
t4g 	General Purpose (Graviton)	 Burstable, good for changing workloads	t is for tiny or turbo
m7g 	General Purpose (Graviton3, ARM)	 Balanced, good for consistent workloads	g is for Graviton (ARM)
m7a 	General Purpose (AMD EPYC, x86)	 Balanced, good for consistent workloads	a is for AMD
c7g 	Compute Optimized (Graviton3)	 High ratio of compute to memory	c is for compute
r7g 	Memory Optimized (Graviton3)	 Good for in-memory databases	r is for RAM
x2gd 	Memory Optimized	 Good for full in-memory apps, local NVMe	x is for xtrime
p5 	Accelerated Computing (H100 GPU)	 Good for AI/ML training	p is for processing (GPU)
i4g 	Storage Optimized (Graviton)	 SSD backed, balance of compute and IOPS	i is for IOPS
d3en 	Storage Optimized	 Highest disk ratio, dense HDD, enhanced networking	d is for dense

EC2 - Instance Type – Current Generation

Type	Family	Description	Mnemonic
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i4g	Storage Optimized (Graviton)	SSD backed, balance of compute and IOPS	i is for IOPS
d3en	Storage Optimized	Highest disk ratio, dense HDD, enhanced networking	d is for dense

- The **instance type** defines the **virtual hardware** supporting an Amazon EC2 instance.
- There are dozens of instance types available, varying in the following dimensions:
 - Virtual CPUs (vCPUs)
 - Memory
 - Storage (size and type)
- Network performance **Instance types** are **grouped into families** based on the **ratio of these values to each other**.
- For instance, the **m4 family** provides a balance of **compute, memory, and network** resources, and it is a good choice for many applications.

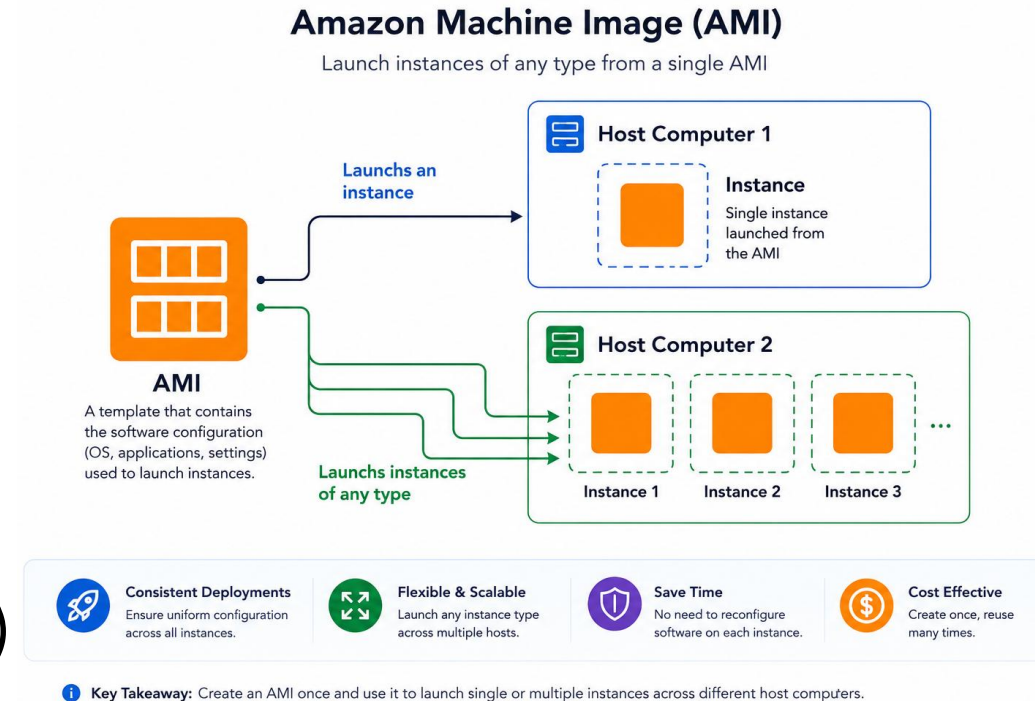
- Within each family there are several choices that scale up linearly in size.
- Note that the ratio of vCPUs to memory is constant as the sizes scale linearly.

EC2 - Instance Type

- Network performance is another key variable when choosing an instance type.
- AWS publishes a relative network performance rating: low, moderate, or high for most types — some instances specify an explicit figure such as 10 Gbps or higher.
- Network performance scales up within a family as the instance size grows.
- For latency-sensitive workloads, AWS offers enhanced networking support (SR-IOV based).

EC2 - Amazon Machine Image (AMI)

- An **AMI** defines the initial **software state** of an **instance** at launch.
- It captures every aspect of that state, including:
 - the **Operating System** and its configuration
 - the initial **patch** level
 - any pre-installed **application** or **system software**
- All standard AMIs are built on x86 (or Arm/Graviton) Linux or Windows operating systems.



EC2 - AMI Types - Summary

4 Ways to Get an Amazon Machine Image (AMI)

Choose the right source for your workload and standardization needs

A

Published by AWS

AWS publishes AMIs across many Linux distributions (Ubuntu, Red Hat, Amazon Linux) and Windows Server versions.

Launching from one gives default OS settings — apply patches immediately after launch, exactly as with a fresh OS install.



AWS Published AMIs



Launch Instance



Apply Patches & Configure



Many OS choices (Linux & Windows)



Quick start with default OS settings



Patch immediately after launch

C

Generated from an existing instance

Launch from a published AMI, configure it to your corporate standard (patches, security, monitoring), then create a new AMI from that instance.

Every future instance inherits the standard — harder for individual projects to drift.



Launch from Published AMI



Configure (Your Standard)



Create New AMI



Launch Future Instances



Consistent & repeatable standards



Includes patches, security, monitoring



Reduces drift across projects

B

AWS Marketplace

An online store where partners publish ready-to-run software images.

Benefits: no manual install, and a license agreement already suited to the cloud.

Cost = standard hourly instance rate + an optional per-hour software fee (some open-source listings add none).



AWS Marketplace



Select Software Image



Launch Instance



Pre-configured with software



License terms integrated for cloud



Cost = instance rate + optional software fee

D

Uploaded virtual servers—

Via AWS VM Import/Export, customers upload images in formats including raw, VMDK, VHD, and OVA.

VMDK = Virtual Machine Disk (VMware/VirtualBox)

VHD = Virtual Hard Disk (Microsoft)

OVF = Open Virtualization Format (XML packaging standard)

OVA = OVF package archived into a single .tar file (OVF + disk files)



Upload Image Files



AWS VM Import/Export



Create AMI



Launch Instance



Bring your existing virtual servers to AWS



Supports RAW, VMDK, VHD, OVA formats



OVF = XML standard
OVA = OVF in a .tar file (single package)

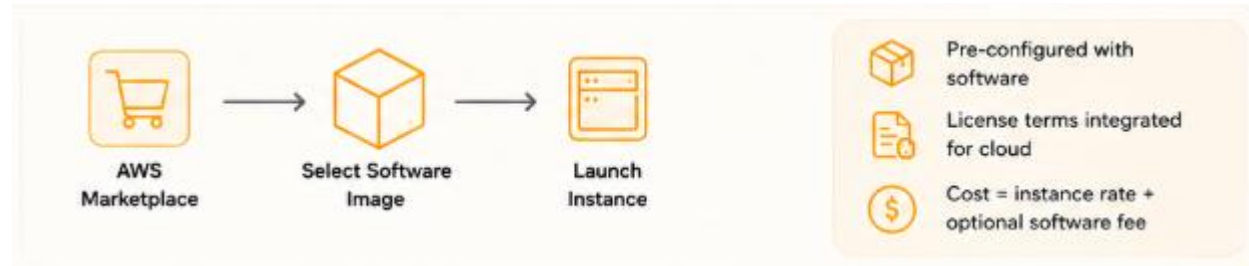
EC2 - AMI Types

- **Published by AWS**— AWS publishes AMIs with versions of many different OSs, both Linux and Windows.
- These include multiple distributions of Linux (including Ubuntu, Red Hat, and Amazon's own distribution) and Windows 2008 and Windows 2012.
- Launching an instance based on one of these AMIs will result in the default OS settings, similar to installing an OS from the standard OS ISO image. As with any OS installation, you should immediately apply all appropriate patches upon launch.



EC2 - AMI Types

- **The AWS Marketplace**— AWS Marketplace is an **online store** that helps **customers find, buy, and immediately start using the software and services that run on Amazon EC2**.
- Many AWS partners have made their software available in the AWS Marketplace.
- This provides two benefits:
 - The customer **does not need to install** the software, and
 - The **license agreement is appropriate** for the cloud.
- Instances launched from an AWS Marketplace AMI incur the standard hourly cost of the instance type plus an additional per-hour charge for the additional software (some open-source AWS Marketplace packages have no additional software charge).



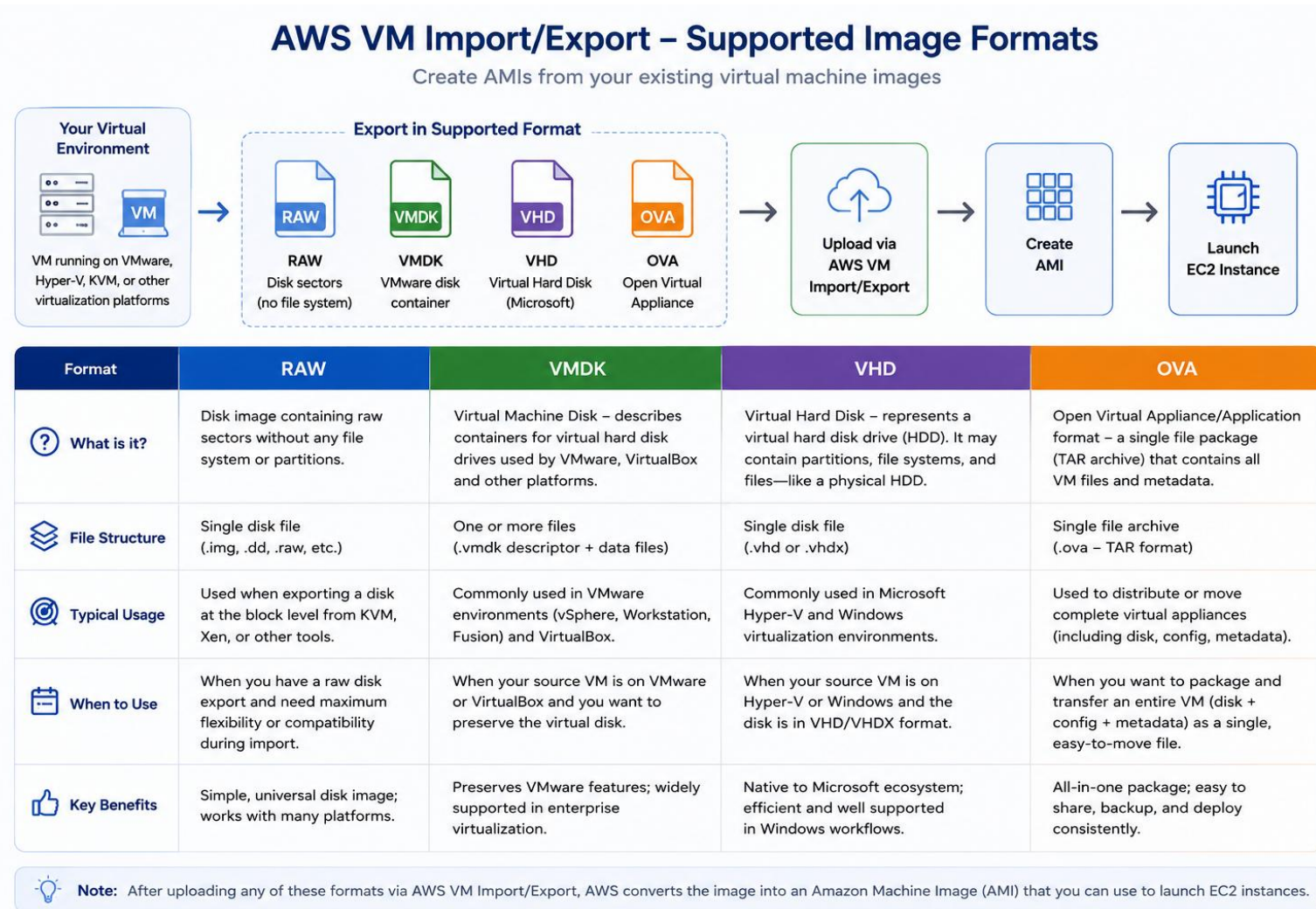
EC2 - AMI Types

- **Generated from Existing Instances** — An AMI can be created from an existing Amazon EC2 instance.
- This is a very **common source** of AMIs. Customers **launch an instance** from a published AMI, and then the **instance is configured to meet all the customer's corporate standards** for updates, management, security, and so on.
- **An AMI is then generated** from the configured instance and used to **generate all instances** of that OS.
- In this way, all new instances **follow the corporate standard** and it is more difficult for individual projects to launch non-conforming instances



EC2 - AMI Types

- **Uploaded Virtual Servers**— Using AWS VM Import/Export service, customers can create images from various virtualization formats, including raw, VHD, VMDK, and OVA. **VMDK** → Virtual Machine Disk : is a file format that describes containers for virtual hard disk drives to be used in virtual machines like VMware Workstation or VirtualBox.
- **VHD** → Virtual Hard Disk: is a file format which represents a **virtual hard disk** drive (HDD). It may contain what is found on a physical HDD, such as disk partitions and a file system, which in turn can contain files and folders. It is typically used as the hard disk of a virtual machine.
- **OVA** → Open Virtual Appliance/Application Format: is merely a single **file** distribution of the same **file** package, stored in the TAR format





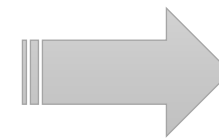
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Connecting to EC2 Instance



Amazon EC2



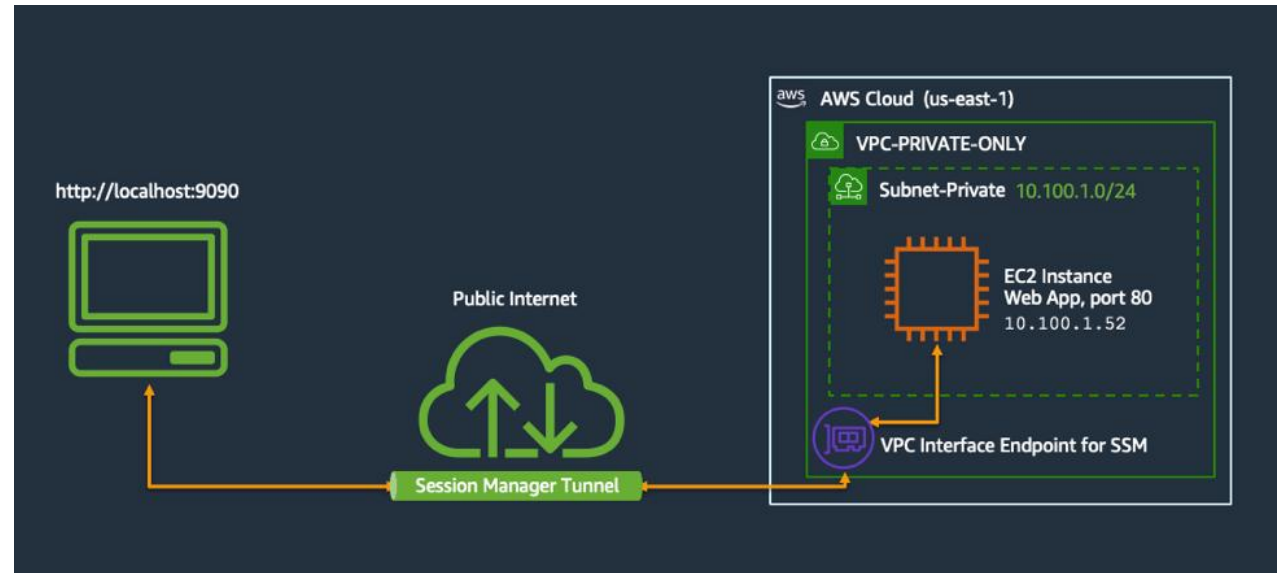
- Virtual Compute Cloud
- Root-level System Access
- Elastic Capacity
- Management API
- Scale in Minutes
- Multiple Instance Sizes
- Network Security Model

Creating an EC2 Instance - Video

- Amazon **EC2** uses **public-key cryptography** to encrypt and decrypt login information.
- **Public-key cryptography** uses a **public key** to **encrypt** a piece of data and an **associated private key** to decrypt the data.
- These two keys together are called a **key pair**.
- **Key pairs** can be created through the **AWS Management Console**, CLI, or API, or customers can upload their own key pairs.
- **AWS stores the public key**, and the **private key** is kept by the **customer**.
- The **private key** is essential to acquiring **secure access** to an **instance** for the **first time**.

EC2 – Accessing over Web

- There are several ways that an instance may be addressed over the web upon creation:
- **Public Domain Name System (DNS) Name**— When you launch an instance, AWS creates a DNS name that can be used to access the instance. This DNS name is generated automatically and cannot be specified by the customer.
- **Public IP**— A launched instance may also have a public IP address assigned. This IP address is assigned from the addresses reserved by AWS and cannot be specified. This IP address is unique on the Internet, persists only while the instance is running, and cannot be transferred to another instance.
- **Elastic IP**— An elastic IP address is a static address unique on the Internet that you reserve independently and associate with an Amazon EC2 instance. EIP's not attached to a running instance are still billed hourly.



EC2 – Lifecycle

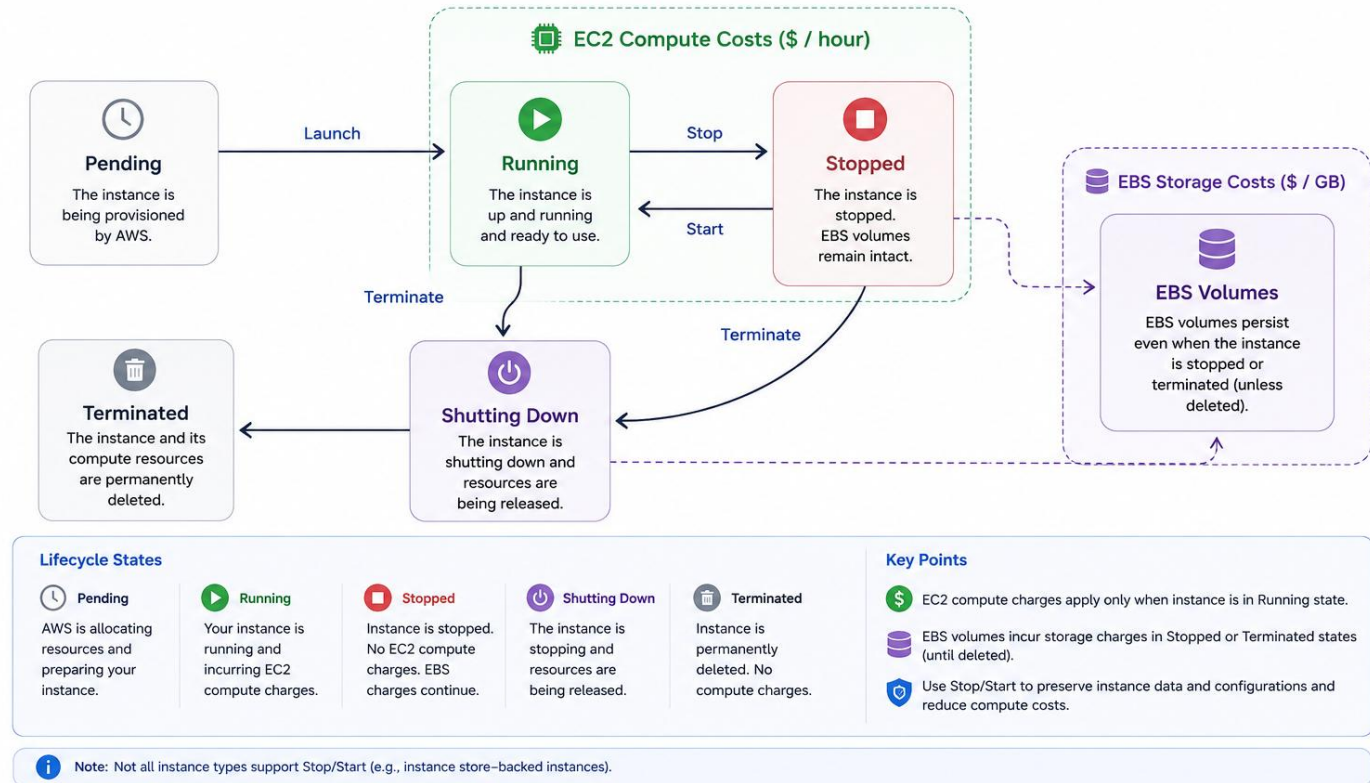
- Amazon EC2 has several features and services that facilitate the management of Amazon EC2 instances over their entire lifecycle.
- Launching → Running → Stopped → Terminated.
- Bootstrapping: The process of providing code to be run on an instance at launch is called bootstrapping.

Managing Instances

- When the number of instances in your account starts to climb, it can become difficult to keep track of them.
- Tags can help you manage not just your Amazon EC2 instances, but also many of your AWS Cloud services.
- Tags are key/ value pairs you can associate with your instance or other service.
- Tags can be used to identify attributes of an instance like project, environment (dev, test, and so on), billable department, and so forth.
- You can apply up to 50 tags per instance.

AWS EC2 Instance Lifecycle

Average lifecycle of an EC2 instance



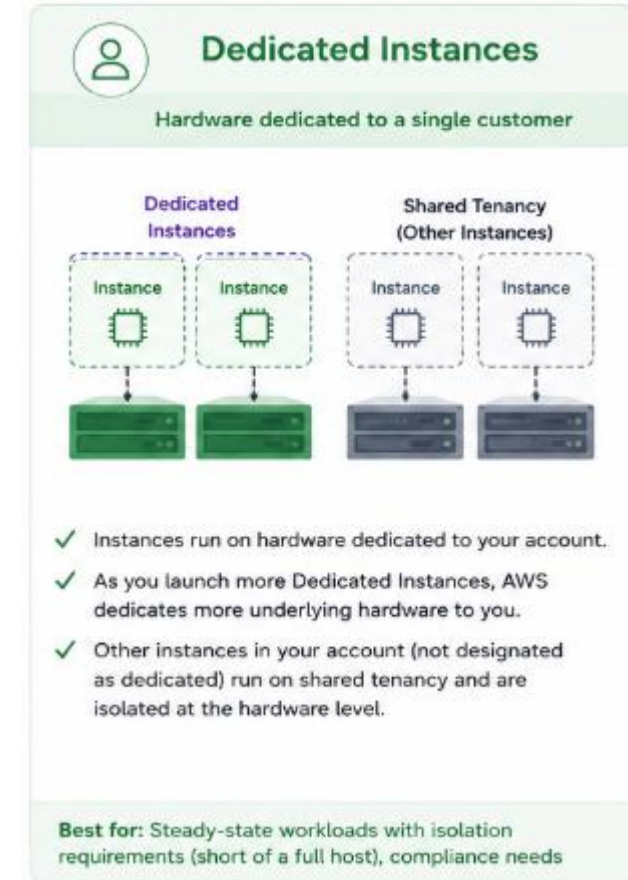
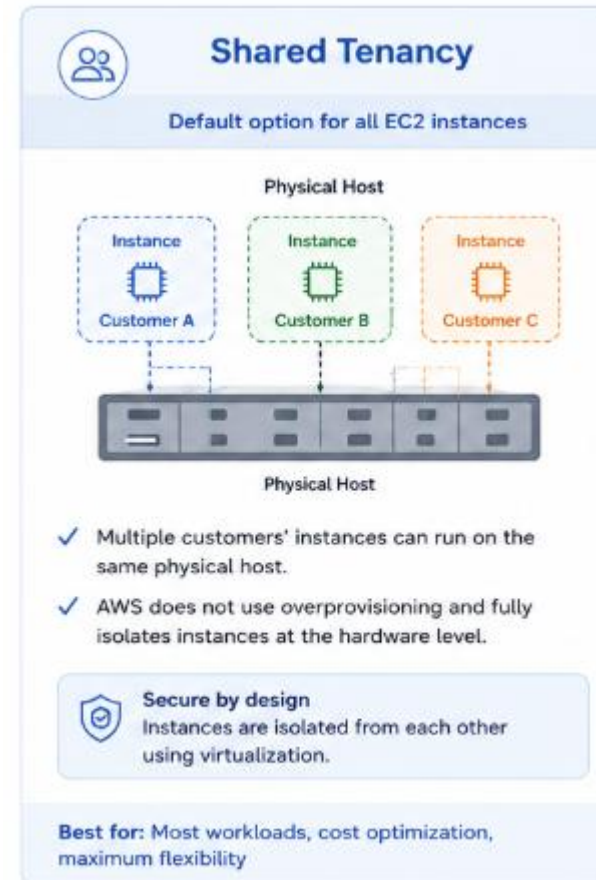
Monitoring Instances

AWS offers a service called Amazon CloudWatch that provides monitoring and alerting for Amazon EC2 instances, and other AWS infrastructure.

EC2 – Tenancy Options

Shared Tenancy Shared tenancy is the default tenancy model for all Amazon EC2 instances, regardless of instance type, pricing model, and so forth. **Shared tenancy means that a single host machine may house instances from different customers.** As AWS does not use overprovisioning and fully isolates instances from other instances on the same host, this is a secure tenancy model.

Dedicated Instances Dedicated Instances run on hardware that's dedicated to a single customer. As a customer runs more Dedicated Instances, more underlying hardware may be dedicated to their account. Other instances in the account (those not designated as dedicated) will run on shared tenancy and will be isolated at the hardware level from the Dedicated



EC2 – Tenancy Options

- **Dedicated Host** An Amazon EC2 Dedicated Host is a physical server with Amazon EC2 instance capacity fully dedicated to a single customer's use. Dedicated Hosts can help you address licensing requirements and reduce costs by allowing you to use your existing server-bound software licenses.



Dedicated Host

Physical server dedicated to a single customer

Dedicated Host



The diagram illustrates a Dedicated Host as a physical server rack. To the right of the rack, four boxes represent EC2 instances, each labeled 'Instance' and containing a microchip icon, indicating that the host is fully dedicated to a single customer's use.

- ✓ A physical server with EC2 instance capacity fully dedicated to your use.
- ✓ Helps address licensing requirements and reduce costs by allowing use of existing server-bound software licenses.
- ✓ You have control over instance placement on the host.

Best for: License compliance, BYOL scenarios, regulatory requirements, maximum isolation

Key Takeaways



All tenancy options provide strong isolation and security by design.



Choose based on your compliance, licensing, and workload consistency needs.

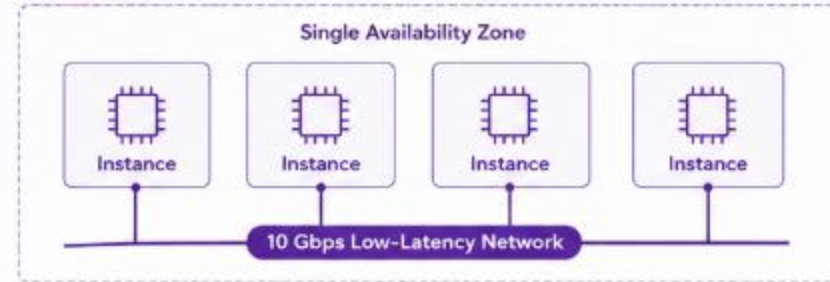


Dedicated options may help optimize costs and simplify license management.

EC2 – Placement Groups

Placement Groups A placement group is a **logical grouping** of instances within a **single Availability Zone**. Placement groups enable **applications to participate** in a low-latency, **10 Gbps network**. To fully use this network performance for your placement group, choose an **instance type** that supports **enhanced networking** and 10 Gbps network performance.

How it works



Common placement group types



Cluster
High performance computing (HPC) workloads



Spread
Critical applications that span racks



Partition
Hadoop, HBase, and other big data applications

Benefits

- ✓ Up to 10 Gbps network bandwidth between instances in the same placement group
- ✓ Lower latency and higher network throughput
- ✓ Improved performance for interdependent workloads



Key Point

Placement groups are logical groupings within a single Availability Zone to provide low-latency, high-bandwidth networking.

EC2 – Instance Store

- **Instance Stores** An instance store (sometimes referred to as **ephemeral storage**) provides **temporary block-level storage** for your instance. This storage is located on disks that are physically attached to the host computer.
- The size and type of **instance stores** available with an Amazon EC2 instance depend on the **instance type**. Can range from **no instance store** to **24 x 2 TB** instance store
- **Instance stores** are included in the cost of an **Amazon EC2 instance**, so they are a very cost-effective solution for appropriate workloads. The key aspect of instance stores is that they are temporary.

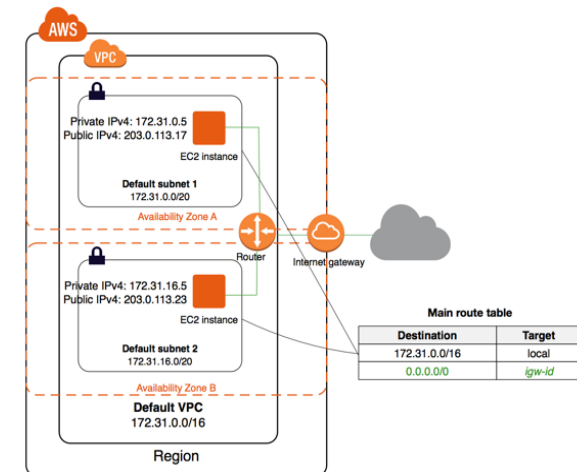


Data in the instance store is lost when:

- The underlying disk drive fails.
- The instance stops (the data will persist if an instance reboots).
- The instance terminates.



Networking in AWS - VPC



AWS VPC



- A **virtual private cloud** (VPC) is a virtual network **dedicated to your AWS account**. It is **logically isolated** from other virtual networks in the AWS cloud.
- You can **launch your AWS resources**, such as Amazon EC2 instances, into **your VPC**. You can provision your own **logically isolated section of AWS**, similar to designing and implementing a separate **independent network** that would operate in an on-premises data center.
- You can configure your VPC; you can select its **IP address range**, **create subnets**, and **configure route tables**, network **gateways**, and **security settings**. A **subnet** is a **range of IP addresses** in your VPC. You can **launch AWS resources into a subnet that you select**.
- Use a **public subnet** for **resources that must be connected to the Internet**, and a **private subnet** for resources that **won't be connected to the Internet**. Within a **region**, you can create **multiple Amazon VPCs**, and each **Amazon VPC** is **logically isolated** even if it shares its IP address space..

uses

- ❖ Build virtual networks on the cloud
- ❖ No need for any VPN, hardware or physical DC
- ❖ Define bespoke network space like:
 - ❖ VPC with a single public subnet only
 - ❖ VPC with public and private subnets
 - ❖ VPC with public and private subnets and AWS Site-to-Site VPN access
 - ❖ VPC with a private subnet only and AWS Site-to-Site VPN access

AWS VPC - Components

WHAT? An Amazon VPC consists of the following components:

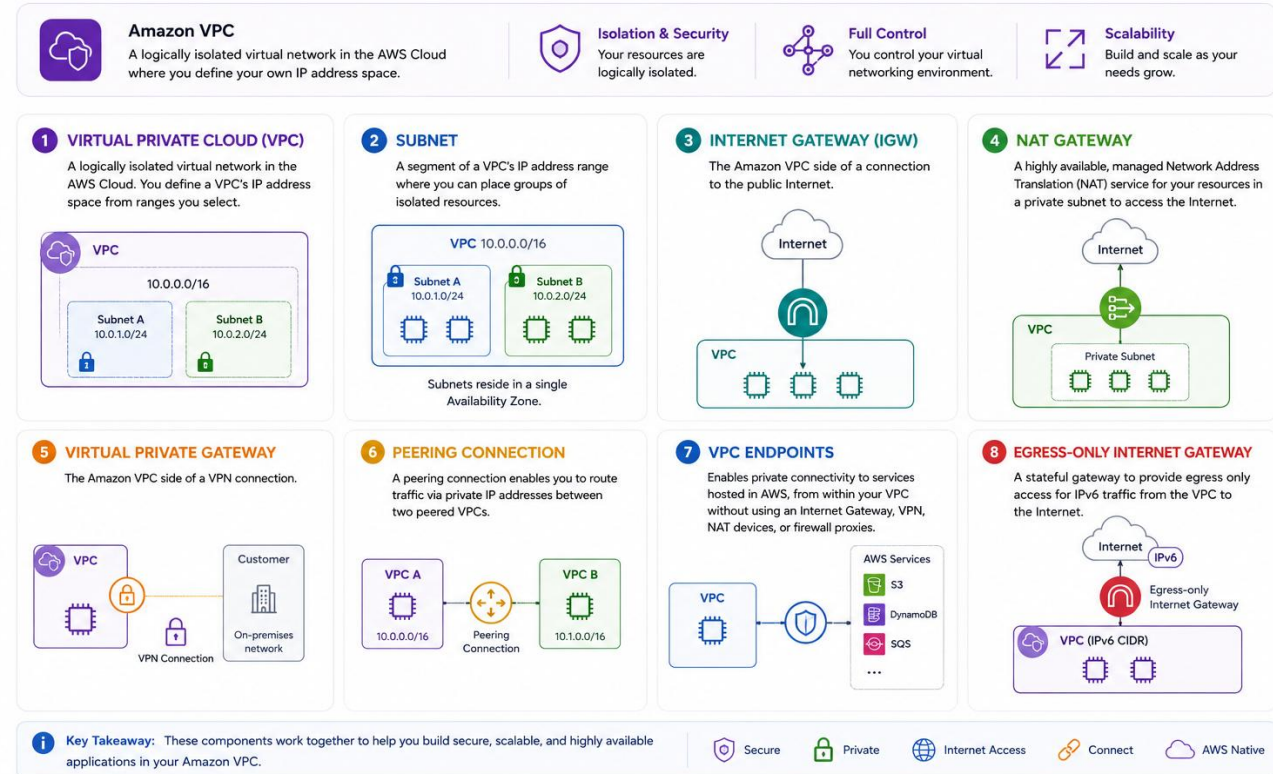
- Subnets
- Route tables
- Dynamic Host Configuration Protocol (DHCP) option sets
- Security groups
- Network Access Control Lists (ACLs)

An Amazon VPC has the following optional components:

- Internet Gateways (IGWs), Elastic IP (EIP) addresses
- Elastic Network Interfaces (ENIs), Endpoints
- Peering
- Network Address Translation (NATs) instances and
- NAT gateways Virtual Private Gateway (VPG), Customer Gateways (CGWs), and Virtual Private Networks (VPNs)

Amazon VPC – Key Networking Components

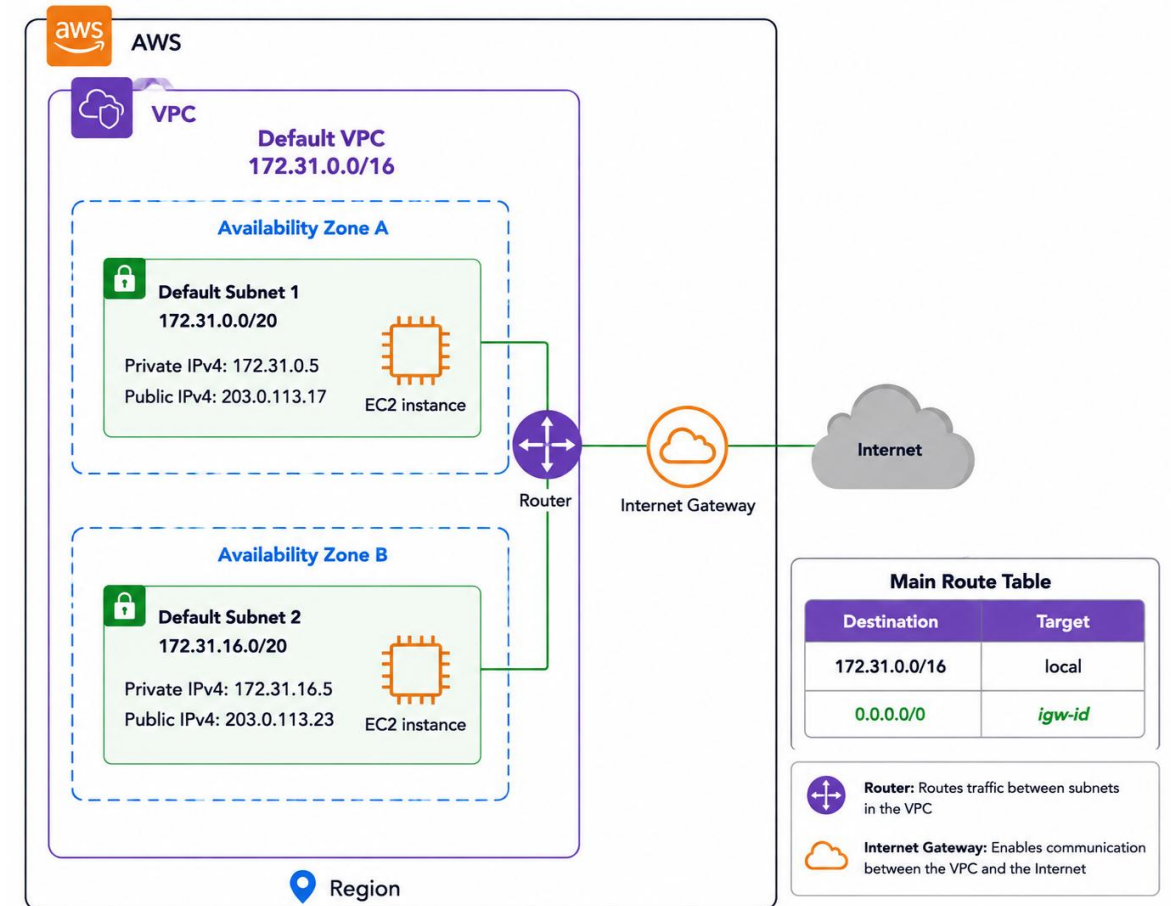
Build logically isolated, secure, and scalable networks in the AWS Cloud



AWS VPC - Functioning



- You control how the instances that you launch into a VPC access resources outside the VPC.
- Your default VPC includes an Internet gateway, and each default subnet is a public subnet.
- Each instance that you launch into a default subnet has a private IPv4 address and a public IPv4 address.
- These instances can communicate with the Internet through the Internet gateway.
- By default, each instance that you launch into a non-default subnet has a private IPv4 address, but no public IPv4 address, unless you specifically assign one at launch, or you modify the subnet's public IP address attribute.
- These instances can communicate with each other, but can't access the Internet.



Creating VPC- Video

Amazon VPC – Key Networking Components

Build logically isolated, secure, and scalable networks in the AWS Cloud



Amazon VPC

A logically isolated virtual network in the AWS Cloud where you define your own IP address space.



Isolation & Security

Your resources are logically isolated.



Full Control

You control your virtual networking environment.



Scalability

Build and scale as your needs grow.

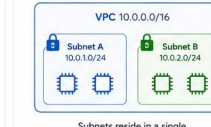
1 VIRTUAL PRIVATE CLOUD (VPC)

A logically isolated virtual network in the AWS Cloud. You define a VPC's IP address space from ranges you select.



2 SUBNET

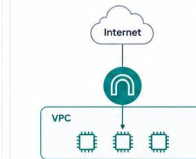
A segment of a VPC's IP address range where you can place groups of isolated resources.



Subnets reside in a single Availability Zone.

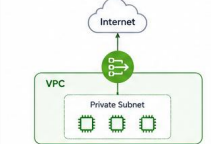
3 INTERNET GATEWAY (IGW)

The Amazon VPC side of a connection to the public Internet.



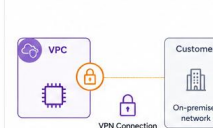
4 NAT GATEWAY

A highly available, managed Network Address Translation (NAT) service for your resources in a private subnet to access the Internet.



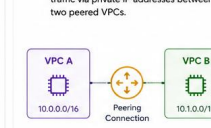
5 VIRTUAL PRIVATE GATEWAY

The Amazon VPC side of a VPN connection.



6 PEERING CONNECTION

A peering connection enables you to route traffic via private IP addresses between two peered VPCs.



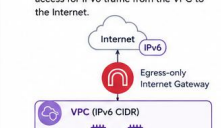
7 VPC ENDPOINTS

Enables private connectivity to services hosted in AWS, from within your VPC without using an Internet Gateway, VPN, NAT devices, or firewall proxies.



8 EGRESS-ONLY INTERNET GATEWAY

A stateful gateway to provide egress only access for IPv6 traffic from the VPC to the Internet.



Key Takeaway: These components work together to help you build secure, scalable, and highly available applications in your Amazon VPC.

Secure Private Internet Access Connect AWS Native



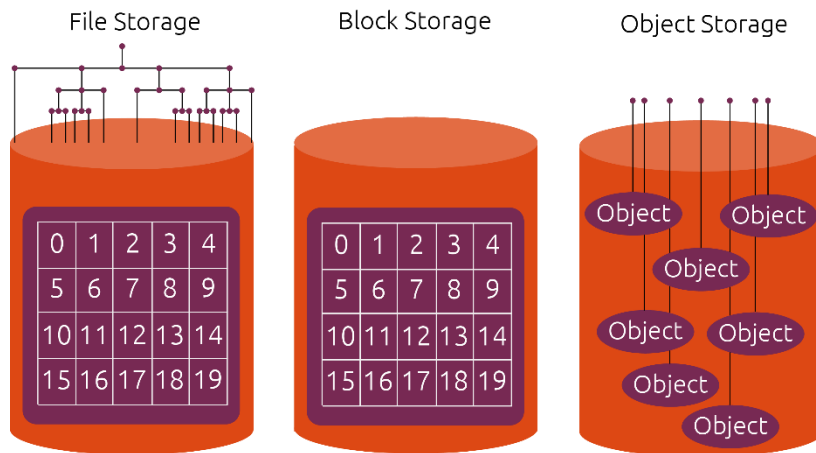
AWS Storage

AWS Cloud Storage Services



Storage Types

- **Block storage** : Operates at a lower level— the raw storage device level— and manages data as a set of numbered, fixed-size blocks.
- **File storage** : Operates at a higher level— the operating system level— and manages data as a named hierarchy of files and folders.
- *Block and file storage are often accessed over a network in the form of a Storage Area Network (SAN) for block storage, using protocols such as iSCSI or Fibre Channel, or as a Network Attached Storage (NAS) file server or “filer” for file storage.*



	Amazon Simple Storage Service (Amazon S3)	A service that provides scalable and highly durable object storage in the cloud.
	Amazon Glacier	A service that provides low-cost highly durable archive storage in the cloud.
	Amazon Elastic File System (Amazon EFS)	A service that provides scalable network file storage for Amazon EC2 instances.
	Amazon Elastic Block Store (Amazon EBS)	A service that provides block storage volumes for Amazon EC2 instances.
	Amazon EC2 Instance Storage	Temporary block storage volumes for Amazon EC2 instances.
	AWS Storage Gateway	An on-premises storage appliance that integrates with cloud storage.
	AWS Snowball	A service that transports large amounts of data to and from the cloud.
	Amazon CloudFront	A service that provides a global content delivery network (CDN).



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AWS Simple Storage Service S3

AWS Cloud Storage Services



AWS S3



- ❖ Amazon S3 is easy-to-use object storage with a simple web service interface that you can use to store and retrieve any amount of data from anywhere on the web.
- ❖ Amazon S3 also allows you to pay only for the storage you actually use, which eliminates the capacity planning and capacity constraints associated with traditional storage.
- ❖ Amazon S3 can be used alone or in conjunction with other AWS services, and it offers a very high level of integration with many other AWS cloud services.

uses

- ❖ Backup and archive for on-premises or cloud data
- ❖ Content, media, and software storage and distribution
- ❖ Big data analytics
- ❖ Static website hosting
- ❖ Cloud-native mobile and Internet application hosting
- ❖ Disaster recovery



AWS S3

Amazon S3 – Core Concepts

Simple, durable, and scalable object storage in the cloud



OBJECTS

The fundamental entities stored in Amazon S3.



- ✓ Opaque data to be stored (1 byte ... 5 Gigabytes)
- ✓ Authentication and access controls



Secure by design: Access controlled by IAM policies, bucket policies, ACLs, and more.



BUCKETS

Containers for objects. Buckets live in a Region.



- ✓ Holds any number of objects
- ✓ Up to 10,000 buckets per account
- ✓ Max object size: **50 TB / object**
Single PUT is capped at 5 TB; larger objects require multipart upload.



Region scoping: Buckets are created in a specific AWS Region.



KEYS

The unique identifier for an object within a bucket.



Object Keys (examples)

- images/photo1.jpg
- videos/intro.mp4
- logs/2024/05/17/app.log
- data/report.csv

- ✓ Unique object identifier within bucket
- ✓ Up to 1024 bytes long
- ✓ Flat object storage model (no folders, just key names with prefixes)



Flat namespace: Use prefixes (e.g., folders/) to organize objects.



Access S3 Anywhere

STANDARDS-BASED INTERFACES

S3 exposes a standards-based API that works with a wide range of tools, libraries, and platforms.

- ✓ **REST API** – Perform all operations using simple HTTP requests
- ✓ **AWS SDK** – Available for all major languages and platforms
- ✓ **URL-Addressability** – Every object has a URL

Applications / Clients



Web App



Mobile App



SDK / CLI

HTTPS

REST API

Amazon S3



- <https://bucket.s3.amazonaws.com/images/photo1.jpg>
- <https://bucket.s3.amazonaws.com/videos/intro.mp4>
- <https://bucket.s3.amazonaws.com/logs/2024/05/17/app.log>
- <https://bucket.s3.amazonaws.com/data/report.csv>



Why it matters



Massive scale
Store unlimited amounts of data



Durability
Designed for 99.99999999% (11 nines) durability



Cost-effective
Pay only for what you store and use



Highly available
Build resilient applications across AWS Regions

Service:

ListAllMyBuckets

Buckets:

CreateBucket

DeleteBucket

ListBucket GetBucketAccessControlPolicy

SetBucketAccessControlPolicy

GetBucketLoggingStatus SetBucketLoggingStatus

Objects:

PutObject

PutObjectInline

GetObject GetObjectExtended

DeleteObject, GetObjectAccessControlPolicy

SetObjectAccessControlPolicy

AWS S3 - Points



- ✓ Amazon S3 is cloud object storage. Instead of being closely associated with a server, Amazon S3 storage is independent of a server and is **accessed** over the **Internet**.
- ✓ Instead of managing data as blocks or files using SCSI, CIFS, or NFS protocols, data is managed as objects using an Application Program Interface **(API) built on standard HTTP verbs**. Each Amazon **S3 object** contains both **data and metadata**.
- ✓ Objects reside in containers called buckets, and each object is identified by a unique user-specified key (filename).
- ✓ Buckets are a simple flat folder with no file system hierarchy.
- ✓ That is, you can have multiple buckets, but you can't have a sub-bucket within a bucket. Each **bucket** can hold an unlimited number of objects.

- ❖ However, keep in mind that **Amazon S3** is not a **traditional file system** and differs in significant ways.
- ❖ In **Amazon S3**, you **GET** an object or **PUT** an **object**, operating on the **whole object** at once, instead of **incrementally updating** portions of the object as you would with a file.
- ❖ Instead of a **file system**, **Amazon S3** is **highly-durable** and **highly-scalable object storage** that is optimized for reads and is built with an intentionally minimalistic feature set.
- ❖ It provides a **simple and robust abstraction** for file storage that frees you from many underlying details that you normally do have to deal with in traditional storage.
- ❖ Amazon S3 objects are automatically replicated on multiple devices in multiple facilities within a region.





AWS Elastic Block Storage EBS

AWS Cloud Storage Services



AWS EBS



- ❖ Amazon EBS provides persistent block-level storage volumes for use with Amazon EC2 instances.
- ❖ Each Amazon EBS volume is automatically replicated within its Availability Zone to protect you from component failure, offering high availability and durability.
- ❖ Amazon EBS volumes are available in a variety of types that differ in performance characteristics and price.
- ❖ Multiple Amazon EBS volumes can be attached to a single Amazon EC2 instance, although a volume can only be attached to a single instance at a time.

uses

- ❖ Boot Volumes
- ❖ SQL & NoSQL Database
- ❖ Big Data workloads
- ❖ Data Warehouses
- ❖ Logging & Telemetry
- ❖ Transaction Processing

AWS EBS Types

General-Purpose SSD General-purpose SSD volumes offer cost-effective storage that is ideal for a broad range of workloads. They deliver strong performance at a moderate price point that is suitable for a wide range of workloads.

A general-purpose SSD volume can range in size from 1 GB to 16 TB and provides a baseline performance of three IOPS per gigabyte provisioned, capping at 10,000 IOPS.

They are suited for a wide range of workloads where the very highest disk performance is not critical, such as:

- System boot volumes
- Small- to medium-sized databases
- Development and test environments



AWS EBS Types

Provisioned IOPS SSD Provisioned IOPS SSD volumes are designed to meet the needs of I/ O-intensive workloads, particularly database workloads that are sensitive to storage performance and consistency in random access I/ O throughput. While they are the most expensive Amazon EBS volume type per gigabyte, they provide the highest performance of any Amazon EBS volume type in a predictable manner.

A Provisioned IOPS SSD volume can range in size from 4 GB to 16 TB. Provisioned IOPS SSD volumes provide predictable, high performance and are well suited for:

- Critical business applications that require sustained IOPS performance
- Large database workloads





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AWS Elastic File Storage EFS

AWS Cloud Storage Services



AWS EFS



- ❖ EFS(Elastic file system) is a file-level storage service that basically provides a shared elastic file system with virtually unlimited scalability support
- ❖ EFS is highly available storage that can be utilized by many servers at the same time. AWS EFS is a fully managed service by amazon and it offers scalability on the fly.
- ❖ This means that the user need not worry about their increasing or decreasing workload. If the workload suddenly becomes higher then the storage will automatically scale itself and if the workload decreases then the storage will itself scale down.
- ❖ This scalability feature of EFS also provides cost benefits as you need not pay anything for the part of storage that you don't use, you only pay for what you use(Utility-based computing).

uses

- ❖ Lift-and-shift application support
- ❖ Analytics for big data
- ❖ Web server support
- ❖ Application development and testing



BBC Uses Shared File Storage to Migrate Red Button Application to the AWS Cloud



Faculty Uses Amazon EFS to Scale Innovative Machine-Learning Platform



T-Mobile Improves Customer Experience, Lowers Costs of Kubernetes Storage Using AWS



Caltech Uses Amazon EFS to Automate Academic Computing File Management



Swisstopo uses EFS to deliver fast map content to millions of users



How one Caribbean university digitally transformed and saved money by migrating to the cloud



Alpha Vertex uses EFS for financial modeling with machine learning



Atlassian uses EFS with Jira Data Center for scalable shared storage



LoanLogics kickstarts its all-in move to the cloud with AWS storage



Be Software simplifies storage and cuts costs using AWS Cloud and EFS



Butterfly Network brings medical imaging to more people using Amazon EFS



Celgene chooses Amazon EFS to store file data for all of its R&D workloads

AWS EFS vs AWS EBS vs AWS S3

AWS Storage Services – Comparison

Choose the right storage service for your workload

Category	 Amazon S3 Object Storage	 Amazon EBS Block Storage	 Amazon EFS File Storage
 Storage Type	Object Storage	Block Storage	File Storage
 Pricing	Pay as you Use	Pay for provisioned capacity	Pay as you Use
 Storage Size	Unlimited Storage	Limited storage (volume size)	Unlimited Storage
 Scalability	Unlimited Scalability	Increase/decrease size manually	Unlimited Scalability
 Durability	Stored redundantly across multiple AZs	Stored redundantly in a Single AZ	Stored redundantly across multiple AZs
 Availability	Max is 99.99% with S3 Standard	99.99%	No SLAs
 Security	Supports Data at Rest and Data in Transit encryption	Supports Data at Rest and Data in Transit encryption	Supports Data at Rest and Data in Transit encryption
 Back up and Restore	Use Versioning or cross-region replication	Automated Backups and Snapshots	EFS to EFS replication
 Performance	Slower than EBS and EFS	Faster than S3 and EFS	Faster than S3, Slower than EBS
 Accessibility	Publicly and Privately accessible	Accessible only via the attached EC2 instance	Accessible simultaneously from multiple EC2 and on-premises instance
 Interface	Web Interface	Block Device Interface	Web and File System Interface (NFS)
 Use Cases	Media, Entertainment, Big data analytics, backups and archives, web serving and content management	Boot volumes, transactional and NoSQL databases, data warehousing ETL	Media, Entertainment, Big data analytics, backups and archives, web serving and content management, home directories



Best for

Unstructured data, data lakes, backup & restore, static website hosting, content delivery



Best for

High performance workloads, databases, boot volumes, low latency applications



Best for

Shared file storage, lift-and-shift applications, content management, home directories



S3: Highly scalable object storage



EBS: High performance block storage for EC2



EFS: Managed file storage for shared access



AZ: Availability Zone



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AWS Glacier

AWS Cloud Storage Services



Amazon Glacier



- Amazon Glacier is an extremely low-cost storage service that provides durable, secure, and flexible storage for data archiving and online backup.
- To keep costs low, Amazon Glacier is designed for infrequently accessed data where a retrieval time of three to five hours is acceptable.
- Amazon Glacier can **store** an **unlimited amount of virtually any kind of data**, in any format.



AWS Glacier

Secure, durable, and ultra-low-cost cloud storage for data archiving and backup



Ultra-Low Cost
Lowest-cost storage in the cloud



Durable & Secure
Designed for 99.999999999% (11 nines) durability and data protection



Built for Archiving
Ideal for long-term data retention and compliance



Fully Managed
No infrastructure management – just store and retrieve



What is AWS Glacier?

AWS Glacier is a secure, durable, and low-cost storage service designed for data archiving and long-term backup. It is ideal for data that is rarely accessed but must be retained for months, years, or even decades.



Storage Classes

Storage Class	Best For	Minimum Storage Duration	Retrieval Options	Typical Retrieval Time
 Glacier Instant Retrieval	Data that needs milliseconds retrieval infrequently	90 days	Instant Retrieval	Milliseconds
 Glacier Flexible Retrieval (formerly Glacier)	Data accessed a few times per year	90 days	- Expedited (1–5 min) - Standard (3–5 hrs) - Bulk (5–12 hrs)	Minutes to Hours
 Glacier Deep Archive	Data rarely accessed (once per year or less)	180 days	- Standard (12 hrs) - Bulk (48 hrs)	Hours to Days



Key Features

- ✓ Highly durable (11 nines durability)
- ✓ Data is automatically replicated within an AWS Region
- ✓ Secure with encryption in transit and at rest
- ✓ Integrates with AWS Identity and Access Management (IAM)



Use Cases

- ✓ Long-term data archiving
- ✓ Regulatory and compliance retention
- ✓ Backup and disaster recovery
- ✓ Medical records, legal documents, and financial records
- ✓ Digital media preservation



Benefits

- ✓ Reduce storage costs significantly
- ✓ Pay only for what you store
- ✓ Scalable – petabytes of data
- ✓ Designed for long-term durability and availability



How it Works

- Upload your data to AWS Glacier
- ↓
- Data is stored securely and redundantly within the region
- ↓
- Retrieve your data when needed using the retrieval options



Security & Compliance

AWS Glacier helps you meet data compliance requirements with secure, durable, and tamper-resistant storage.



Data encrypted in transit and at rest

Note: Retrieval times are typical and may vary. Minimum storage duration charges apply if data is deleted before the minimum period.



AWS Database Services

AWS Cloud Storage Services



AWS Database Services

Review: AWS Managed Database Services

Purpose-built, managed databases for every application need



Amazon DynamoDB

Key-value database



Serverless, highly scalable
Single-digit millisecond latency at any scale



Built-in security
Encryption at rest and in transit, fine-grained access control



Fully managed
No servers to provision, patch, or manage



Best for

- ✓ High-throughput applications
- ✓ Mobile, web, gaming backends
- ✓ Shopping carts, user profiles
- ✓ IoT, real-time analytics



Amazon RDS

Relational database (SQL)



Managed relational databases
Supports MySQL, PostgreSQL, MariaDB, Oracle, SQL Server



Automated operations
Backups, patching, monitoring, and failover



Scalable & available
Read replicas, Multi-AZ deployments



Best for

- ✓ Web applications
- ✓ E-commerce platforms
- ✓ SaaS applications
- ✓ Traditional business apps



Amazon ElastiCache

In-memory data store



Blazing fast performance
Microsecond latency with in-memory caching



Supports Redis & Memcached
Open-source, widely adopted engines



High availability
Multi-AZ with auto-failover and replication



Best for

- ✓ Caching layers
- ✓ Session stores
- ✓ Leaderboards, counters
- ✓ Real-time analytics



Amazon Redshift

Data warehouse



Petabyte-scale analytics
Fast analytics on large datasets



Columnar storage
Optimized for complex queries and aggregations



Built for the cloud
Massively parallel processing (MPP) architecture



Best for

- ✓ Data warehousing
- ✓ Business intelligence (BI)
- ✓ Reporting & dashboards
- ✓ Large-scale analytics



Fully Managed by AWS

Automated backups, monitoring, patching, scaling, and high availability so you can focus on your applications.



Secure

Built-in security and compliance



High Performance

Low latency and high throughput



Scalable

Scale up or down in minutes



Cost Effective

Pay only for what you use

AWS Managed Database Services – At a Glance

Choose the right database for your application

Service	 Amazon DynamoDB Key-value Database	 Amazon RDS Relational Database (SQL)	 Amazon ElastiCache In-memory Data Store	 Amazon Redshift Data Warehouse
 What it is	A fully managed NoSQL key-value and document database.	A managed relational database service that supports popular SQL engines.	An in-memory data store and cache service for sub-millisecond latency.	A fully managed, petabyte-scale data warehouse for analytics.
 Key Characteristics	• Serverless • Single-digit millisecond latency • Schema-less (flexible data model) • Built-in security • Highly durable and available	• Supports MySQL, PostgreSQL, MariaDB, Oracle, SQL Server • ACID compliant • Automated backups and patching • Read replicas & Multi-AZ	• Microsecond latency • Supports Redis & Memcached • In-memory caching • High throughput	• Columnar storage • Optimized for analytical queries • Massively parallel processing (MPP) • Integrates with data lake & BI tools
 Performance	Single-digit millisecond latency at any scale	Milliseconds (varies by engine and workload)	Microsecond latency	Optimized for high-throughput analytical queries (seconds to minutes for complex queries)
 Scalability	Virtually unlimited (automatic scaling)	Scale up (vertical) or read scale out with read replicas	Scale out with read replicas (Redis) and clustering (Redis or Memcached)	Scale out (add compute nodes) to petabytes
 Security	Encryption at rest & in transit, IAM fine-grained access control	Encryption, VPC, IAM, security groups, compliance certifications	Encryption in transit & at rest (Redis), VPC, IAM	Encryption, VPC, IAM, column/row level security, compliance certifications
 Pricing Model	Pay-per-request or on-demand capacity	Pay for provisioned instances (by instance type & storage)	Pay for nodes (cache instances) and backup storage	Pay for compute nodes and storage (RA3 managed storage)
 Primary Uses	• Mobile apps & gaming backends • Shopping carts & user sessions • IoT & real-time analytics • High-throughput workloads	• Web applications • E-commerce platforms • SaaS applications • Traditional business applications	• Caching layers • Session stores • Leaderboards & counters • Real-time analytics	• Data warehousing • Business intelligence (BI) • Reporting & dashboards • Large-scale analytics
 Best For	High scale, flexible schema, low-latency key-value workloads	Relational data, transactions, and complex queries	Low-latency caching and real-time data access	Large datasets, complex analytics, and business intelligence
 How to Choose?	Need ultra-low latency key-value access at scale? Choose DynamoDB .		Need relational data with ACID compliance and SQL? Choose RDS .	
			Need microsecond speed for caching or sessions? Choose ElastiCache .	
			Need to analyze large volumes of data? Choose Redshift .	

AWS RDS



- ❖ Amazon RDS is a service that simplifies the setup, operations, and scaling of a relational database on AWS
- ❖ Amazon RDS supports seven database engines: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2, and Amazon Aurora (its own MySQL/PostgreSQL-compatible engine).
- ❖ Multi-AZ deployments maintain a synchronous standby in a second AZ for automatic failover — this is a durability/availability feature, not a performance feature.
- ❖ Read Replicas are asynchronous, read-only copies (same-region or cross-region) used specifically to scale read-heavy workloads — up to 5 for most engines, up to 15 for Aurora.
- ❖ Automated backups (default: on) retain 1–35 days of history, enabling point-in-time recovery to the second.

uses

- ❖ Lift-and-shift application support
- ❖ Analytics for big data
- ❖ Web server support
- ❖ Application development and testing



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AWS Case Study



Zomato



Who & what

Zomato — India's restaurant-discovery and food-delivery platform, serving 70M+ daily users across 24 countries — runs its analytics data platform (Apache Trino and Druid) on AWS.



What they did

Migrated Trino and Druid clusters from x86 EC2 instances to AWS Graviton2-based (Arm) EC2 instances, using EC2 Spot Instances to further cut compute cost.



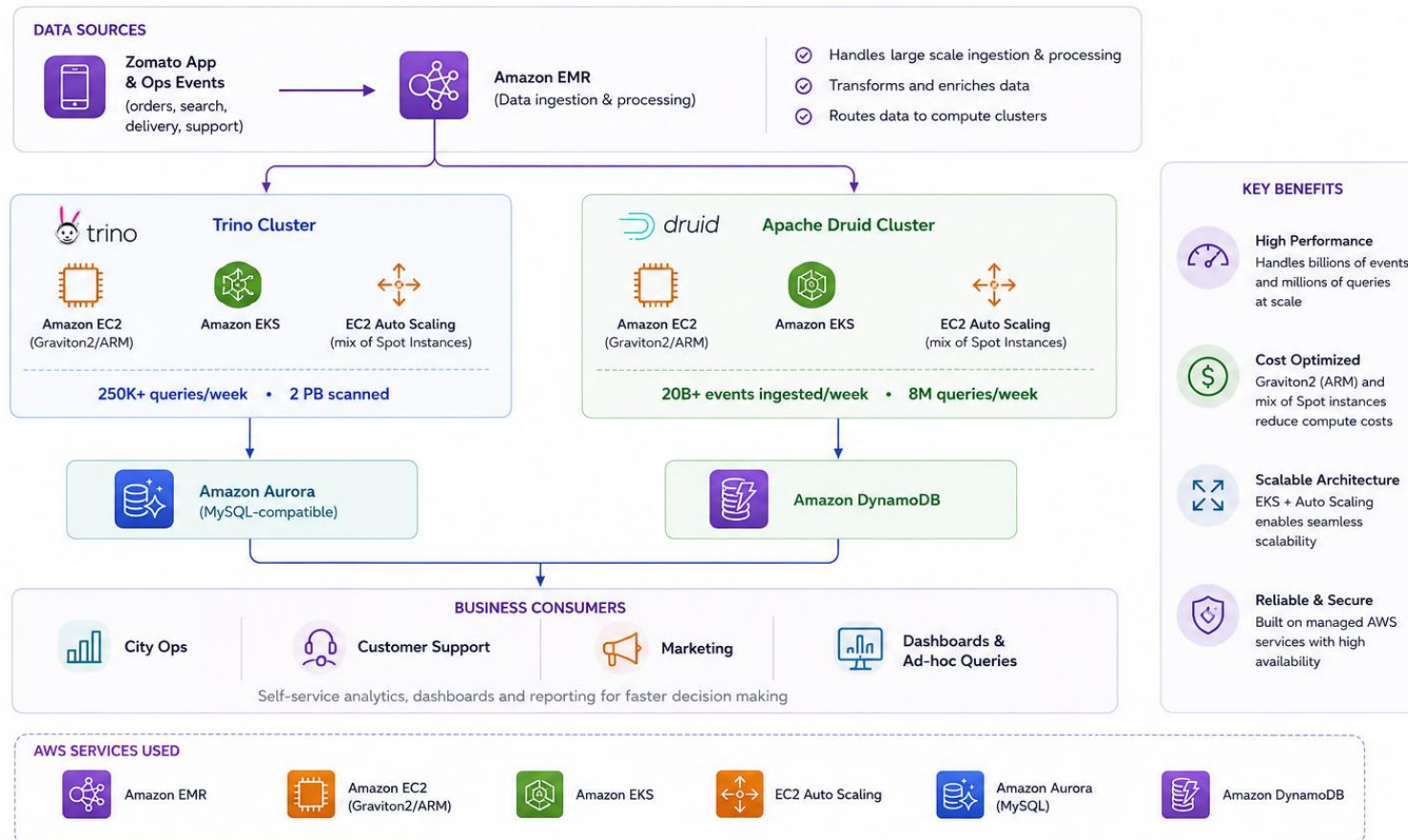
Result (AWS-published)

30% reduction in compute cost, 25% improvement in query performance, and roughly \$300K/year saved on the Trino cluster alone.



Case Study – Zomato Data Platform Architecture

High-scale data platform on AWS powering analytics and operational insights



Source: AWS Open Source Blog, "How Zomato Boosted Performance 25% and Cut Compute Cost 30% Migrating Trino and Druid Workloads to AWS Graviton."

Note: Architecture simplified for clarity. Components and layout inferred for teaching purposes.

Q & A.....





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Credits

*Hwang, Kai; Dongarra, Jack; Fox, Geoffrey C.. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things (Kindle Locations 3532-3533). Elsevier Science. Kindle Edition.